

```

-----
# Step 1: Import libraries
# -----
import pandas as pd
from mlxtend.preprocessing import TransactionEncoder
from mlxtend.frequent_patterns import apriori, association_rules

# -----
# Step 2: Load datasets
# -----
orders = pd.read_csv('/content/order_products__prior.csv')
products = pd.read_csv('/content/products.csv')

# -----
# Step 3: Merge datasets
# -----
df = orders.merge(products, on='product_id')

# -----
# Step 4: Prepare data
# -----
df = df[['order_id', 'product_name']]
df.columns = ['OrderID', 'Item']

print("Sample Data:\n", df.head())

# -----
# Step 5: Convert to transactions
# -----
transactions = df.groupby('OrderID')['Item'].apply(list).tolist()

print("Total Transactions:", len(transactions))

# -----
# Step 6: Sampling (VERY IMPORTANT)
# -----
transactions = transactions[:2000] # reduce size for speed

# -----
# Step 7: Encoding
# -----
te = TransactionEncoder()
te_data = te.fit(transactions).transform(transactions)
df_encoded = pd.DataFrame(te_data, columns=te.columns_)

print("Encoded Shape:", df_encoded.shape)

# -----
# Step 8: Apply Apriori
# -----
frequent_itemsets = apriori(df_encoded, min_support=0.01, use_colnames=True)

print("\nFrequent Itemsets:\n", frequent_itemsets.head())

# -----
# Step 9: Generate Rules11111111111111111111
# -----
rules = association_rules(frequent_itemsets, metric="confidence", min_threshold=0.3)

rules = rules.sort_values(by='confidence', ascending=False)

print("\nTop Association Rules:\n")
print(rules[['antecedents', 'consequents', 'support', 'confidence', 'lift']].head())

```

```

return datetime.utcnow().replace(tzinfo=utc)
/usr/local/lib/python3.12/dist-packages/jupyter_client/session.py:203: DeprecationWarning: datetime.datetime.utcnow() is d
return datetime.utcnow().replace(tzinfo=utc)
/usr/local/lib/python3.12/dist-packages/jupyter_client/session.py:203: DeprecationWarning: datetime.datetime.utcnow() is d
return datetime.utcnow().replace(tzinfo=utc)
/usr/local/lib/python3.12/dist-packages/jupyter_client/session.py:203: DeprecationWarning: datetime.datetime.utcnow() is d
return datetime.utcnow().replace(tzinfo=utc)

Frequent Itemsets:
  support      itemsets
0  0.0105      (100% Raw Coconut Water)
1  0.0155      (100% Whole Wheat Bread)
2  0.0130      (2% Reduced Fat Milk)
3  0.0260      (Apple Honeycrisp Organic)
4  0.0185      (Asparagus)

Top Association Rules:

      antecedents      consequents  support  confidence  \
4  (Organic Fuji Apple)      (Banana)    0.0145    0.446154
1      (Cucumber Kirby)      (Banana)    0.0110    0.407407
2      (Honeycrisp Apple)      (Banana)    0.0120    0.387097
0 (Organic Raspberries) (Bag of Organic Bananas)    0.0140    0.325581
3      (Organic Avocado)      (Banana)    0.0160    0.307692

      lift
4  2.814851
1  2.570394
2  2.442251
0  2.782747
3  1.941276
/usr/local/lib/python3.12/dist-packages/jupyter_client/session.py:203: DeprecationWarning: datetime.datetime.utcnow() is d
return datetime.utcnow().replace(tzinfo=utc)
/usr/local/lib/python3.12/dist-packages/jupyter_client/session.py:203: DeprecationWarning: datetime.datetime.utcnow() is d
return datetime.utcnow().replace(tzinfo=utc)
/usr/local/lib/python3.12/dist-packages/jupyter_client/session.py:203: DeprecationWarning: datetime.datetime.utcnow() is d
return datetime.utcnow().replace(tzinfo=utc)
/usr/local/lib/python3.12/dist-packages/jupyter_client/session.py:203: DeprecationWarning: datetime.datetime.utcnow() is d
return datetime.utcnow().replace(tzinfo=utc)

```